CSCI3170
Introduction to Database Systems

8_normal_forms_intro
Slides: Michael Yu

Normal Forms

- Purpose:
  - Serve an standards for “good” database design
  - (i.e., resolve data anomalies due to redundancy)
  - Formalized by functional (or other) dependencies

- Types of Normal Forms:
  - 1NF, 2NF, 3NF (Codd 1972)
  - Boyce-Codd NF (1974)

First Normal Form (1NF)

- 1NF simply asserts that attribute values be atomic.
  - (Relational model definition asserts at least 1NF)

CRS_PREF			
Prof	Course	Fac_Dept	Crs_Dept
Smith	353	Comp Sci	Comp Sci
Smith	379	Comp Sci	Comp Sci
Smith	221	Comp Sci	Decision Sci
Clark	353	Comp Sci	Comp Sci
Clark	351	Comp Sci	Comp Sci
Clark	379	Comp Sci	Comp Sci
Clark	456	Comp Sci	Mathematics
Turner	353	Chemistry	Comp Sci
Turner	456	Chemistry	Mathematics
Turner	272	Chemistry	Chemistry
Jamieson	353	Mathematics	Comp Sci
Jamieson	379	Mathematics	Comp Sci
Jamieson	221	Mathematics	Decision Sci
Jamieson	456	Mathematics	Mathematics
Jamieson	469	Mathematics	Mathematics

(Recap) First Normal Form (1NF)

- 1NF relation **drawbacks**(1) **Repetition** of data!

- **Example (see CRS_PREF below):**
  - Fact that (given professor in given department) may have to repeat
  - Fact that (given course offered by given department) may have to repeat

- No structural protection against insertion and update anomalies, etc.,

CRS_PREF			
Prof	Course	Fac_Dept	Crs_Dept
Smith	353	Comp Sci	Comp Sci
Smith	221	Comp Sci	Decision Sci
Turner	353	Chemistry	Comp Sci
Turner	456	Chemistry	Mathematics

(Recap) First Normal Form (1NF)

- 1NF relation **drawbacks**: (2) **Some associations not explicitly stored!**

- **Example (see CRS_PREF below):**
  - Association between professor and department not recorded unless professor has some course references
  - Association between course and department not recorded unless someone prefers course

- No structural protection against deletion anomalies, etc.,

CRS_PREF			
Prof	Course	Fac_Dept	Crs_Dept
Smith	353	Comp Sci	Comp Sci
Smith	379	Comp Sci	Comp Sci
Turner	353	Chemistry	Comp Sci

Second Normal Form (2NF) (1)

- **2NF Prerequisite (1)**

  - (Prime attribute): Attribute of relation schema R is called a prime attribute of R if it is a member of **some** candidate key of R.
  - (Nonprime attribute): Attribute is called nonprime if it is not a prime attribute—that is, if it is not a member of **any** candidate key.

- *Q1:* ***Why all keys rather than just one key?***
  Answer: No special consideration will be given to the primary key over other candidate keys.

- *Q1.5:* ***Can we consider just the primary key instead?***
  Answer: Only if the relation has one candidate key only

Second Normal Form (2NF) (2)

- **2NF Prerequisite (2)**

  - (Full functional dependency): In an FD X→ Y , Y is fully functionally dependent on X if there is **no** Z ⊂ X such that Z → Y .
  - (Partial functional dependency): In an FD X→ Y , Y is partially functionally dependent on X if there is **any** Z ⊂ X such that Z → Y .

- *Q2:* ***Is it possible for a nonprime attribute to be partially dependent on the key?***
  Answer: Yes

Example

- *CRS_PREF(Prof, Course, Fac_Dept, Crs_Dept)*
  - *F = {*
    Prof → Fac_Dept,
    Course → Crs_Dept
    }

- From FDs, we can see…
  - only one candidate key: *{Prof, Course}*
  - Fac_Dept depends only on Prof and not on Course
  - Crs_Dept depends only on Course and not on Prof

CRS_PREF			
Prof	**Course**	**Fac_Dept**	**Crs_Dept**
Smith	353	Comp Sci	Comp Sci
Smith	379	Comp Sci	Comp Sci
Smith	221	Comp Sci	Decision Sci
Clark	353	Comp Sci	Comp Sci
Clark	351	Comp Sci	Comp Sci
Clark	379	Comp Sci	Comp Sci
Clark	456	Comp Sci	Mathematics
Turner	353	Chemistry	Comp Sci
Turner	456	Chemistry	Mathematics
Turner	272	Chemistry	Chemistry
Jamieson	353	Mathematics	Comp Sci
Jamieson	379	Mathematics	Comp Sci
Jamieson	221	Mathematics	Decision Sci
Jamieson	456	Mathematics	Mathematics
Jamieson	469	Mathematics	Mathematics

Second Normal Form (2NF) (3)

- (Full functional dependency): In an FD X→ Y , Y is fully functionally dependent on X if there is no Z ⊂ X such that Z → Y.

- (Second Normal Form): A relation schema *R* is in second normal form (2NF) if **every** *nonprime* attribute *A* in *R* is *not partially dependent* on ***any*** key of *R*.

- Point:
  - 2NF be first NF to consider restrictions on the normal form

Second Normal Form (2NF)

F = {
Prof → Fac_Dept,
Course → Crs_Dept
}

- Let's decompose this 1NF relation:

CRS_PREF			
Prof	**Course**	**Fac_Dept**	**Crs_Dept**
Smith	353	Comp Sci	Comp Sci
Smith	379	Comp Sci	Comp Sci
Smith	221	Comp Sci	Decision Sci
Clark	353	Comp Sci	Comp Sci
Clark	351	Comp Sci	Comp Sci
Clark	379	Comp Sci	Comp Sci
Clark	456	Comp Sci	Mathematics
Turner	353	Chemistry	Comp Sci
Turner	456	Chemistry	Mathematics
Turner	272	Chemistry	Chemistry
Jamieson	353	Mathematics	Comp Sci
Jamieson	379	Mathematics	Comp Sci
Jamieson	221	Mathematics	Decision Sci
Jamieson	456	Mathematics	Mathematics
Jamieson	469	Mathematics	Mathematics

Second Normal Form (2NF)

F = {
Prof → Fac_Dept,
Course → Crs_Dept
}

- (You can decompose a 1NF into a 2NF)

COURSE_PREF	
Prof	**Course**
Smith	353
Smith	379
Smith	221
Clark	353
Clark	351
Clark	379
Clark	456
Turner	353
Turner	456
Turner	272
Jamieson	353
Jamieson	379
Jamieson	221
Jamieson	456
Jamieson	469

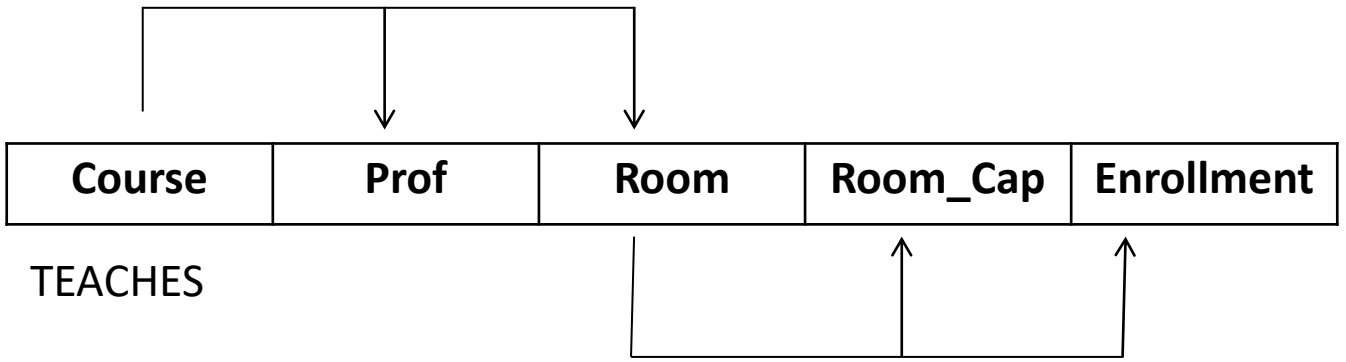
COURSE	
Course	**Crs_Dept**
353	Comp Sci
379	Comp Sci
221	Decision Sci
351	Comp Sci
456	Mathematics
272	Chemistry
469	Mathematics

FACULTY	
Prof	**Fac_Dept**
Smith	Comp Sci
Clark	Comp Sci
Turner	Chemistry
Jamieson	Mathematics

Second Normal Form (2NF)

- (Second Normal Form): A relation schema *R* is in second normal form (2NF) if every nonprime attribute *A* in *R* is not partially dependent on *any* key of *R*.

- Example: Is this table 2NF? (Yes)



Redundancy/Issue in 2NF

- Do we resolve all issues? Not for all relations in 2NF.
  1. The fact that A532 has Room_Cap of 45 and Enrol_Lmt of 40 can still be stored twice.
  2. If course 355 is deleted, then the fact that H940 has Room_Cap of 400 and Enrol_Lmt of 300 will be lost.

TEACHES				
Course	**Prof**	**Room**	**Room_Cap**	**Enrollment**
353	Smith	*A532*	*45*	*40*
351	Smith	C320	100	60
355	*Clark*	*H940*	*400*	*300*
456	Turner	B278	50	45
459	Jamieson	D110	50	45
500	*Bob*	*A532*	*45*	*40*

Third Normal Form (3NF) (1)

- **3NF Prerequisite (1)** (superkey): A superkey of a relation schema $R = \{A1, A2, \ldots, An\}$ is a set of attributes $S \rightarrow R$ with the property that no two tuples $t1$ and $t2$ in any legal relation state r of R will have $t1[S] = t2[S]$.

- **3NF Prerequisite (2)** (key): A key K is a superkey with the additional property that removal of any attribute from K will cause K not to be a superkey anymore.

Third Normal Form (3NF) (2)

- Recall (Second Normal Form): A relation schema *R* is in second normal form (2NF) if every nonprime attribute *A* in *R* is not partially dependent on *any* key of *R*.

- Definition (Third Normal Form):
  A relation scheme is in third normal form (3NF) if for all non-trivial FD's of the form

$$X \rightarrow A$$

  - Either
    X is a superkey
  - or
    A is a prime attribute.

Third Normal Form (3NF) (3)

- (Third Normal Form): A relation scheme is in third normal form (3NF) if for all non-trivial FD's of the form X → A. Either X is a superkey or A is a prime attribute.

- Let us enumerate all options:

X	**A**
Superkey	Prime
Superkey	Nonprime
Not Superkey	Prime
Not Superkey	Nonprime

  - **3NF disallows FDs of the form "Not superkey -> Nonprime"**
  - Note: Not Superkey = either (1) *prime attributes that are proper subset of a key* or (2) *nonprime*

Third Normal Form (3NF) (4)

- Compare 2NF and 3NF

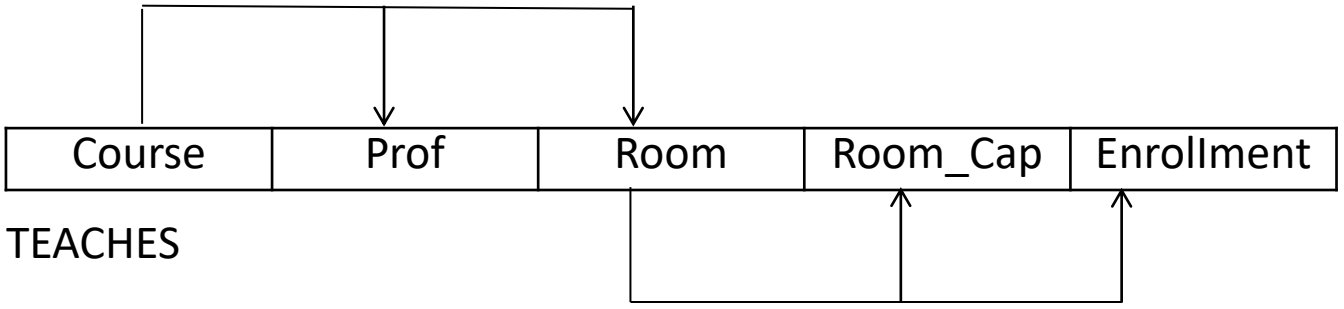
<table>
<tr><th colspan="2">X</th><th>A</th></tr>
<tr><td colspan="2">Superkey</td><td>Prime</td></tr>
<tr><td colspan="2">Superkey</td><td>Nonprime</td></tr>
<tr><td colspan="2">Not Superkey</td><td>Prime</td></tr>
<tr><td rowspan="2">Not Super Key</td><td>Prime attributes (proper subsets)</td><td rowspan="2">Nonprime</td></tr>
<tr><td>Non prime</td></tr>
</table>

  - 2NF allows FDs of the type nonprime -> nonprime
  - 3NF disallows nonprime -> nonprime in addition

Third Normal Form (3NF) (5)

- 3NF disallows transitive dependencies of non-prime attributes on a candidate key
  - (Transitive dependency):
    - A FD X→Y is a transitive dependency if there is a Z that is not a subset of **any candidate** key, such that X → Z and Z → Y.
    - We say that attributes of Y are transitively dependent on X

- Example (See TEACHES below):
  - TEACHES has only one candidate key.
  - Hint: any transitive dependency of a nonkey attribute on our candidate key?



Third Normal Form (3NF) (6)

- We can decompose TEACHES from 2NF into 3NF:

TEACHES				
Course	**Prof**	**Room**	**Room_Cap**	**Enrollment**
353	Smith	*A532*	*45*	*40*
351	Smith	C320	100	60
355	*Clark*	*H940*	*400*	*300*
456	Turner	B278	50	45
459	Jamieson	D110	50	45
500	*Bob*	*A532*	*45*	*40*

2NF
(but not 3NF)

COURSE_DETAILS		
Course	**Prof**	**Room**
353	Smith	A532
351	Smith	C320
456	Turner	B278
459	Jamieson	D110
355	Clark	H940

ROOM_DETAILS		
Room	**Room_Cap**	**Enrollment**
A532	45	40
C320	100	60
B278	50	45
D110	50	45
H940	400	300

3NF

Summary

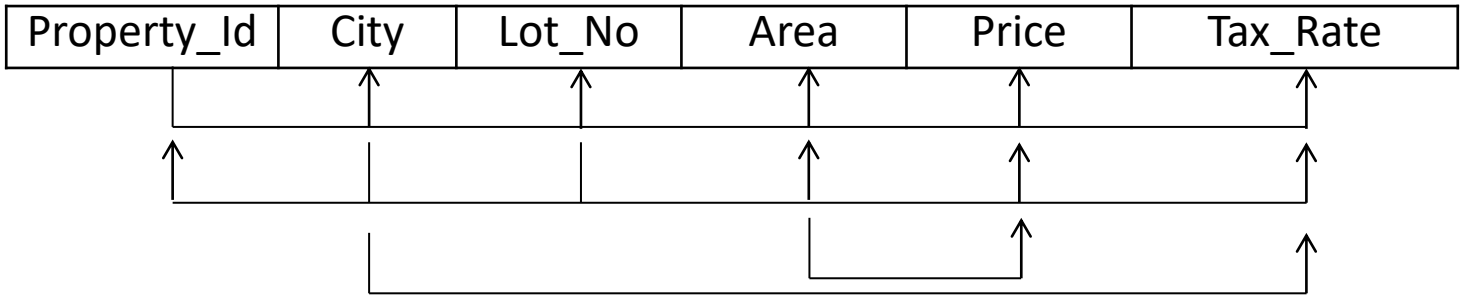
- We have covered 1NF, 2NF, 3NF (Codd 1972)

- Details:
  - Each normal form satisfies requirements of lower normal forms
  - Normal form of a relation is the highest normal form it satisfies.
  - A relation is “normalized“ if it meets 3NF

- ***"The key, the whole key, and nothing but the key, so help me Codd”***

Review

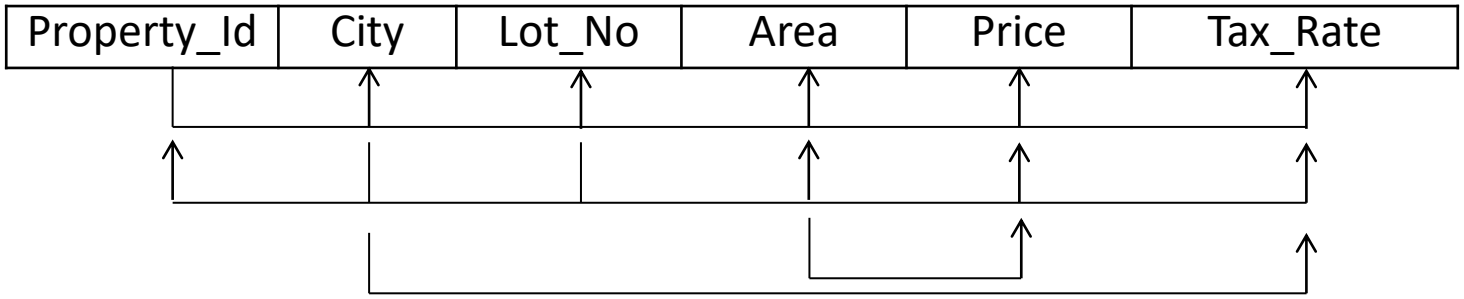
- Scenario: LOTS



- What normal form is this?
  - (Suppose given that this satisfies 1NF)
  - Is it in 2NF?

Review

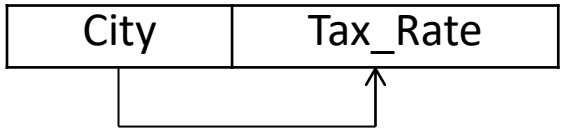
- Scenario: LOTS



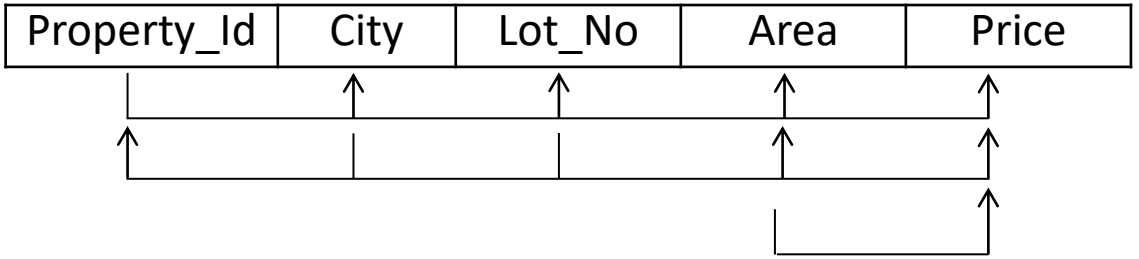
- What normal form is this?
  - LOTS is **not in 2NF**: *Since City→Tax_Rate, Tax_Rate* is not prime, and *{City,Lot_No}* is a key, making Tax_Rate partially dependent on a key.

Review

- Update! LOTS now 2NF compliant
  - LOTS1



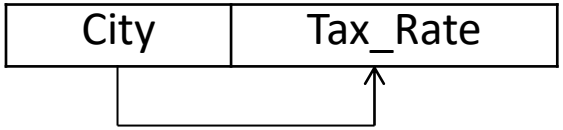
  - LOTS2



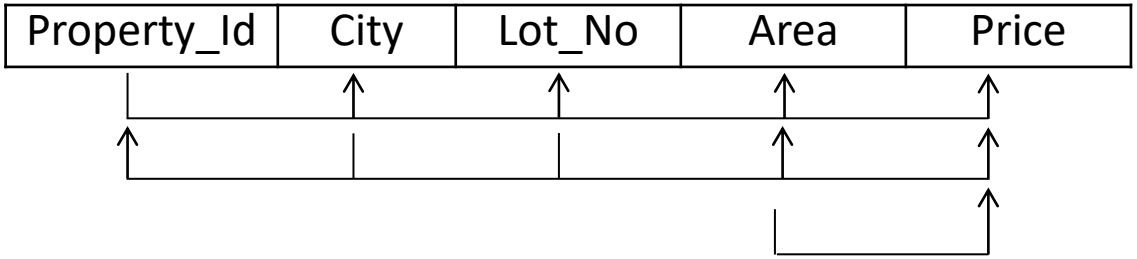
- Could this in 3NF?

Review

- Update! LOTS now 2NF compliant
  - LOTS1



  - LOTS2



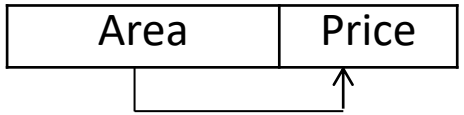
- Could this in 3NF?
  - LOTS2 is **not in 3NF**: Since Area → Price, {Area} is not a superkey and Price is not prime.
  - *"A **transitive dependency** in Third Normal Form (3NF) happens when a non-prime attribute (a column that is not part of any candidate key) depends on another non-prime attribute rather than directly on the primary key"*

Third Normal Form (3NF) (7)

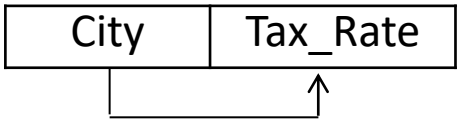
- LOTS1A



- LOTS1B

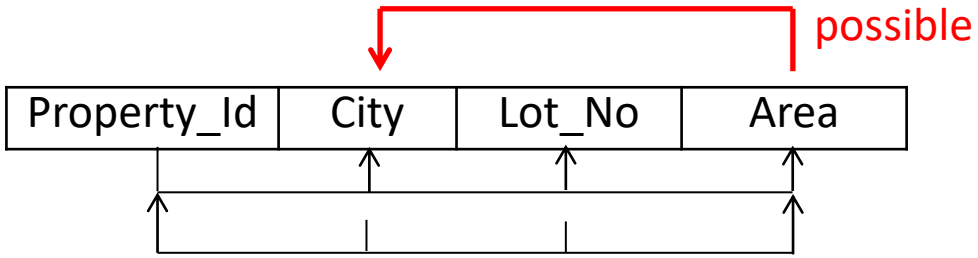


LOTS2

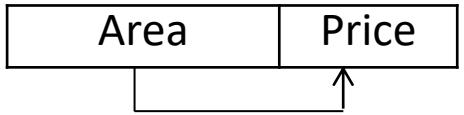


Third Normal Form (3NF) (8)

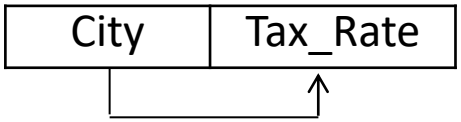
- LOTS1A



- LOTS1B



LOTS2



- It is possible that dependency *Area* → *City* exists
- Relation schemes are still in 3NF since City is a prime attribute. However, there can be anomalies, just as before.

3NF Example

- Consider a BOOKING relation *R*
  - *R = (title, theater, city)*
    F = {
    theater → city
    title, city → theater
    }

- Two candidate keys: *{title, city}* and *{theater, title}*
  - R is in 3NF
  - *Title, city → theater*
    {title, city} is a superkey
  - *theater → city*
    city is contained in a candidate key

Redundancy in 3NF

- There is still some **redundancy** to be found in this schema

- Example of problems due to redundancy in 3NF
  - R = (title, theater, city)
  - F = {
    theater $\rightarrow$ city,
    Title, city $\rightarrow$ theater
    }

title	*theater*	*city*
j_1	l_1	k_1
j_2	l_1	k_1
j_3	l_1	k_1

- Repetition of information (e.g., the relationship l_1, k_1)

Boyce-Codd Normal Form (BCNF) (1)

- Definition (Third Normal Form):
  A relation scheme is in third normal form (3NF) if for all non-trivial FD's of the form X → A
  - Either X is a superkey
  - or A is a prime attribute.

- Definition (Boyce-Codd Normal Form):
  - A relation scheme is in Boyce-Codd Normal Form (BCNF) if whenever X→A holds and X→A is non-trivial, X is a superkey.
  - also known as 3.5NF

Boyce-Codd Normal Form (BCNF)

- Definition (Boyce-Codd Normal Form): A relation scheme is in Boyce-Codd Normal Form (BCNF) if whenever X→A holds and X→A is non-trivial, X is a superkey.
- Let us enumerate all options:

X	A
Superkey	Prime
Superkey	Nonprime
~~Not Superkey~~	~~Prime~~
~~Not Superkey~~	~~Nonprime~~

- Note: not Superkey = either (1) *prime attributes that are proper subset of a key* or (2) *nonprime*

Comparisons

- BCNF implies 3NF

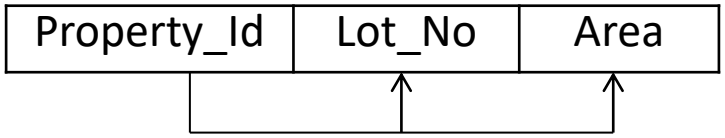
Property	3NF	BCNF
Elimination of redundancy due to functional dependency	Most	Yes
Lossless Join	Yes	Yes
Dependency preservation due to functional dependency	Yes	Maybe

- It is not always possible to get a BCNF decomposition that is dependency preserving (dependency preservation discussed in future lectures)

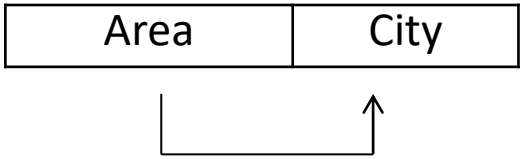
Boyce-Codd Normal Form (BCNF)(cont)

- We can make our example into BCNF:

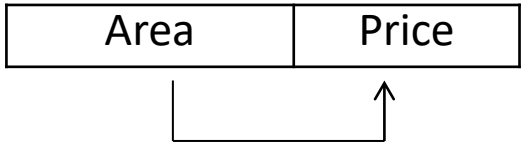
LOTS1AA



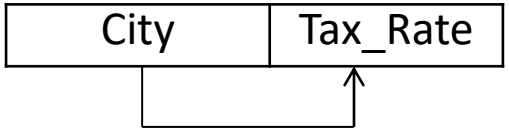
LOTS1AB



LOTS1B



LOTS2

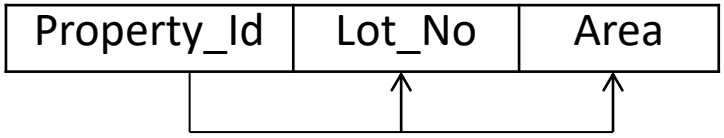


- (One possible decomposition to satisfy BCNF)

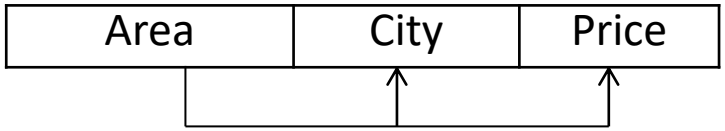
Boyce-Codd Normal Form (BCNF)(cont)

- We can make our example into BCNF:

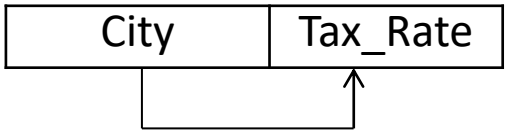
LOTS1AA



LOTS1C



LOTS2



- (LOTS1AB and LOTS1B (from prev slide) into LOTS1C)

Discussion

- We discussed that the normal form of a relation is the highest NF it satisfies. E.g., R is 2NF means that R is not 3NF or BCNF.

- In terms of the database scheme as a whole…
  - A database scheme is in 1NF if all its relations are in 1NF.
  - A database scheme is in 2NF if all its relations are in 2NF.
  - A database scheme is in 3NF if all its relations are in 3NF.
  - A database scheme is in BCNF if all its relations are in BCNF.

Learning Outcomes

- Definitions of 1NF, 2NF, 3NF and BCNF
- (3NF and BCNF is the key NFs to understand)
- How the presence of some types of functional dependencies can contribute to update anomalies
- Determine the highest NF of a relation